

| PROFILE STATEMENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
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| Ph.D. student in Computer Science at William & Mary, advised by Prof. Yifan Sun. My academic focus is computer architecture, accelerator simulation, and hardware/software co-design. I have a background in electrical and computer engineering, digital design, and computer systems. My current work is part of my doctoral training and academic research at William & Mary.                                                                                                                                                                         |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| EDUCATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| Doctor of Philosophy in Computer Science<br>College of William & Mary, , Williamsburg, VA<br>Advisor: Prof. Yifan Sun<br>GPA:4.0/4.0                                                                                                                                                                                                                                                                                                                                                                                                                     | Aug. 2024 - Present                                                                                                                                                                      |                                                                                                                                                                                                                  |
| Master of Science in Electric and Computer Engineering<br>Georgia Tech, Atlanta, Atlanta, GA<br>GPA: 3.55/4.0                                                                                                                                                                                                                                                                                                                                                                                                                                            | Aug. 2021 - May 2023                                                                                                                                                                     |                                                                                                                                                                                                                  |
| Bachelor of Science in Computer Engineering Chip – Scale Integration<br>Virginia Tech, Blacksburg, VA<br>GPA: 3.36/4.0                                                                                                                                                                                                                                                                                                                                                                                                                                   | Aug. 2017 - May 2021                                                                                                                                                                     |                                                                                                                                                                                                                  |
| CORE SKILLS AND EXPERTISE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"><li>Digital Design</li><li>Digital Testing</li><li>Computer Organization &amp; Architecture</li><li>Circuit Design</li><li>Hardware security</li></ul>                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"><li>Software Development</li><li>Academic Research</li><li>Robotics</li><li>Embedded System Design</li><li>DC/AC Electrical Circuit Analysis</li></ul> | <ul style="list-style-type: none"><li>Technical Project Management</li><li>Data Structure and Algorithms</li><li>Technical Analysis</li><li>Machine Learning</li><li>Problem Analysis &amp; Resolution</li></ul> |
| ACADEMIC AND TECHNICAL SKILLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <b>Programming:</b> C/C++, Verilog, Python, TCL, Matlab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <b>Software:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"><li><b>EDA tools:</b> Cadence Virtuoso, Siemens Tessent, Verdi, Synopsys Design Compiler, Synopsys PrimeTime, Synopsys Formality</li><li><b>FPGA tools:</b> Quartus, Modelsim</li></ul>                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <b>Hardware:</b> FPGAs, Arduino, ARM-based embedded systems, TI-MCU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <b>Platform:</b> Windows, Mac, Linux (Ubuntu&CentOS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <b>Communication:</b> Design proposals, technical reports, presentations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| WORKING EXPERIENCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <b>DFT Engineer</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <b>GWX Technology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| July 2023 – April 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"><li>Designed and implemented digital circuit DFT strategies, including SCAN/MBIST/JTAG and IP tests, ensuring efficient and effective testing processes</li><li>Conducted thorough pre-simulation, post-simulation and verification of DFT and test patterns, ensuring accurate and reliable results</li><li>Collaborated with team members to develop and optimize ATPG designs</li><li>Assisted in ATE(automatic test equipment) debugging, troubleshooting any issues that arose during the testing phase</li></ul> |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| TECHNICAL PROJECT HIGHLIGHTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                          |                                                                                                                                                                                                                  |
| <b>A High-Performance Spatial Architecture Simulator - William&amp;Mary, Williamsburg, VA</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                          | Sep 2024 – Present                                                                                                                                                                                               |
| <ul style="list-style-type: none"><li>Developed research infrastructure for modeling and analyzing execution behavior in spatial architectures</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                          |                                                                                                                                                                                                                  |

- Designed intermediate representations and data-processing pipelines to capture program mapping, communication, and execution metadata.
- Implemented simulation components for scheduling, resource tracking, interconnect behavior, and runtime performance analysis.
- Built validation, debugging, and profiling tools to support functional checking, trace inspection, and experiment comparison.

#### **A High-Performance FPGA SoC – GWX Inc., Shenzhen, Guangdong, China**

**Sep 2023 – April 2024**

- Joining in developing a full testing system for a high-performance FPGA SoC
- Responsible for partial block's scan chain insertion using Tessent DFT Flow and Synopsys DFT flow on Linux platform
- Implementing SSN technique in Tessent DFT flow to optimize the whole DFT design
- Successfully increasing the atpg coverage for stuck fault from 98.56% to 99.43% in pre-simulation
- Responsible for building whole simulation environment from pre-sim to post-sim including BSCAN, MBIST, and ATPG
- Communicating with front-end and back-end engineers
- Expected to be taped out in the end of April 2024

#### **SRAM Design – Georgia Tech, Atlanta, GA**

**Aug 2022 – Dec 2022**

- Designed a complete system of a 16-rows by 32-columns SRAM array and peripherals
- Compute expected size of transistors and design the schematics of SRAM cells
- Extract the netlist and write the HSPICE code to do simulations for schematics
- Draw the layout of SRAM array and do extraction-based simulations

#### **Robot Rover Developed Project – Virginia Tech, Blacksburg, VA**

**Feb 2021 – May 2021**

- Successfully designed a new prototype of robot rover and applied it to practical utilization
- Built an effective working relationship with team members
- Wrote C algorithm code for the sensor component such as IR sensor and TOF lidar sensor
- Enhanced the sensor measurement accuracy and aided the navigation system
- Used Ti board to build a computation and communication system with MQTT protocols

#### **Systolic Array Chip VLSI Design Project – Virginia Tech, Blacksburg, VA**

**Aug 2020 – Dec 2020**

- Designed a 4x4 systolic array system including peripherals
- Designed a processing element including a carry look-ahead adder
- Documented concepts and feature requirements for development by the team
- Designed basic circuit of processing element with peripheral system and implemented layout to the circuit

#### **Hokster 8-bit processor Developed Project – Virginia Tech, Blacksburg, VA**

**Jan 2020 – Apr 2020**

- Utilized strong technical expertise to design a new Harvard architecture 8-bit softcore processor
- Professionally developed applications for testing using python and assembly language
- Designed a reduced instruction set computer with room for future expansion
- Enhanced performance testing efficiency and streamlined testbench code by reducing the line count by 20%

### **PUBLICATIONS**

- Rezvani, B., Conroy, T., Beckwith, L., Bozzay, M., Laffoon, T., McFeeters, D., **Shi, Y.**, Vu, M., & Diehl, W. (2020). Efficient Simultaneous Deployment of Multiple Lightweight Authenticated Ciphers. Cryptology ePrint Archive, Paper 2020/609. <https://eprint.iacr.org/2020/609>